Fall 2025 ECEN 629 Homework Set 1

Due Sep. 16 in class, or by Sep. 16 12:45 pm on Canvas.

Prob. 1. Compute the norms of the following vector $x = (-100, -99, -98, \ldots, 99, 100)^T$.

(a) ℓ_1 norm;

(b) ℓ_∞ norm;

(c) ℓ_2 norm.

Note: Please consult Appendix A.1.2 if you are not familiar with the concepts.

Prob. 2. For a quadratic function $f(x) = x^T Px + q^T x + r$ where P is not necessarily symmetric, show that the gradient is $\nabla f(x) = (P + P^T)x + q$. Find the Hessian of the function. Provide details in your proofs.

Note: Please consult Appendix A.4 if you are not familiar with the concepts.

Prob. 3. A matrix $A \in \mathbf{R}^{m\times n}$ can also be viewed as a vector, by stacking all its n columns into a single length-mn column. Let $x \in \mathbf{R}^m, y \in \mathbf{R}^n$ be two column vectors. Show that

$$\mathrm{tr}(A^\top xy^\top) = \langle xy^T, A\rangle = x^T Ay,$$

where tr is the trace of the matrix.

Prob. 4. Let $B \in \mathbf{R}^{n\times n}$ be a given symmetric matrix of size n-by-n. Determine if the following set is convex: the set of all n-by-n symmetric A matrices, such that $A - B$ is positive semidefinite. If it is convex, provide your reasoning; if not, provide a counter-example.

Prob. 5. A set is defined as

$$\left\{x \in \mathbf{R}^n \middle| ||x - a||_2 \le \theta ||x - y||_2 \text{ for all } y \in S\right\}$$

where $S \subseteq \mathbf{R}^n$.

(a) Determine whether the set is convex, which may depend on the value of θ. For the case that it is, provide your reasoning; otherwise, provide a counter-example.

(b) Let $a = (1, 0)^T$ and $S = \{(-1, 0)^T, (0, -1)^T\}$. For $\theta = 2$ and $\theta = 0.5$, sketch the resultant set, respectively.

Prob. 6. (a) Show that if S_1 and S_2 are convex set in $\mathbf{R}^n$, then the set sum

$$S \triangleq \{x_1 + x_2 | x_1 \in S_1, x_2 \in S_2\}$$

is also convex.

(b) Let $S_1 = \{(-1, -1)^T\}$ and $S_2 = \left\{(2, 2)^T + u \middle| ||u|| \le 1\right\}$. What is their set sum?

(c) Show that if S_1 and S_2 are convex set in $\mathbf{R}^n$, then the inverse sum

$$S \triangleq \bigcup_{1\ge\lambda\ge 0} \{\lambda S_1 \cap (1 - \lambda)S_2\}$$

is also convex, where $\lambda S_j = \{\lambda x_j | x_j \in S_j\}$, $j = 1, 2$.

(d) Let $S_1 = \{(1,1)^T\}$ and $S_2 = \left\{(2,2)^T + u \middle| ||u|| \leq 1\right\}$. What is their inverse sum?

Prob. 7. It is known that one halfspace contains another in the following form

$$\{x | a^T x \leq b\} \subseteq \{x | c \cdot a^T x \leq \tilde{b}\}.$$

Provide the conditions that the parameters c, b, and $\tilde{b}$ must satisfy. Are these conditions also sufficient for the containment relation between these two halfspaces?

Note: As noted in class, the two sets are halfspaces; therefore, they must satisfy the definition of a halfspace.

Prob. 8. For a pair of given sets $C, D \subseteq \mathbf{R}^n$, consider the following set

$$S = \left\{(a,b) \in \mathbf{R}^{n+1} \middle| a^T x \leq b \text{ for all } x \in C, a^T x \geq b \text{ for all } x \in D\right\}.$$

Prove that the set S is convex.

Prob. 9. Let $C = \{X \in \mathbb{R}^{n \times n} \mid a^T X a \geq 0, \forall a \in \mathbb{R}^n\}$. Determine whether the set

$$\mathcal{S} = \{A \in \mathbb{R}^{n \times n} \mid A - M \in C\}$$

is convex, where $M \in C$ is a fixed matrix. Provide justification or a counterexample. Note that usually the definition of a matrix being positive-definite requires the matrix to be symmetric, but this problem removes that condition.

Prob. 10. Plot the norm balls of radius $r = 1$ for the following norms in $\mathbb{R}^2$ and $\mathbb{R}^3$:

(a) ℓ_1 norm;

(b) ℓ_∞ norm;

(c) ℓ_2 norm.